

Artificial Intelligence and the Racial Gap in Mortgage Lending

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What I like: the right question at the right time — theory (Fuster et al. 2022) and lab evidence (Bowen et al. 2025) say “this *could* happen,” and this paper goes to the field. A heavy data lift (Revelio $\times$ HMDA $\times$ McDash), and it anticipates alternative channels.

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1. Measure each bank-year's gap **within** the bank-year
2. Regress those gaps on Bank AI **across** bank-years

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$\Rightarrow$ **Identifying assumption: absent AI, high- and low-AI banks would have the same minority gap**

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Suggestion 3: show step 2 — bin-scatter $\hat{g}_{b,t}$ against *Bank AI*

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- **Bhutta, Hizmo & Ringo (JF 2025):** confidential HMDA credit scores and race-blind government AUS recommendations explains *most* of the denial gap
- Table 13 has credit score (for approved loans), but sample difference is large (e.g. minority coefficient moves substantially)

Suggestion: add credit score to **Table 7** — you already have the McDash merge. You test CLTV and LTI; FICO is the one that matters.

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- $AI\ Strong = 1$ if **at least one** employee in 2008–10 graduated from an AI-strong university
 - Mechanically increasing in headcount — a bank with 200k employees surely qualifies
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Suggestions: continuous exposure + *log 2008–10 headcount* $\times$ *Minority* as a control; report robust F and Anderson–Rubin CIs; plot % *AI Strong* by size decile.

Picture I'd love: reduced form *AI Strong* $\times$ *Minority* by year, 2005–2024.

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Suggestion 3: split the postings — credit/risk roles vs. everything else

(Time permitting) Pricing: two reasons to lead with the rate spread

1. **Timing.** 2022 rates went $\sim 3\% \rightarrow \sim 7\%$ *within the year*
 - Year FE cannot absorb the lock date
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 - Minority and white borrowers face *identical* point-rate schedules, but sort to different points
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Side-step both: make **rate spread (APR – APOR)** the headline pricing outcome. APOR is benchmarked at lock (timing ✓) and APR includes points and fees (sorting ✓). You have it: Table 12, col. (4), $1.782^{**} \approx 3.6$ bp/SD.

(Time permitting) Heterogeneity: my eyeball tests — and gaps vs. levels

Contrast	Coefficients	z (diff)	Verdict
Not sold vs. GSE (rate)	2.446 vs. 0.079	2.65	Different ✓
Purchase vs. refi (denial)	0.413 vs. 0.209	0.93	Can't reject equal
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“Unequal” vs. “harmful.” Table 2, col. (1): at AI banks denials *fall* for *both* groups (-0.71 vs. -0.18 p.p. per SD). That is exactly Fuster et al.’s *predictably unequal* — minorities benefit *less*.

⇒ Report outcomes by group in levels; the welfare story differs.

Summary

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A lot to like, and many threads to pull. **Looking forward to the next draft!**

Thanks!

Notes for the author
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Background: I mean this as a full conveyance of information, as I would to a grad school friend. Please take anything below in the spirit in which it is given... Friends in the pursuit of knowledge. Caveat/disclosure: This is also augmented with observations from an LLM. This is comments from my reading excluding those already mentioned above.

Sample and measurement

- **The denial variable is internally inconsistent.** §6.1 says withdrawn/incomplete applications are “treated as being denied,” but Table 1 shows $N = 13.9\text{M}$ for *Denial* vs. 19.0M for *Withdrawn or incomplete*, and Table 11 col. (2) has 13.4M . A **sample-flow table** would help (purposes, lien, occupancy, 1–4 family, preapprovals, action codes, drops at each step). If withdrawals *are* in *Denial*, the underwriting-only estimate (0.417 , Table 11) is the better headline.
- §2.3 says the sample requires non-missing interest rates — that cannot hold for the denial sample. Say which filters apply to which sample.
- A 20.1% denial rate is high vs. the usual HMDA purchase/refi figures — points back to what is in the sample.
- **Small denominators:** 5 postings with 1 AI listing $\Rightarrow \text{Bank AI} = 20\%$. Add a minimum-postings threshold or weight by postings; also report $\log(\# \text{ AI postings})$ and an any-AI indicator.

Specification and inference

- Does $Bank\ AI = 0$ mean “no AI hiring” or “thin LinkedIn coverage”? The second is measurement error correlated with size.
- Report the *within-bank* share of $Bank\ AI$'s variance (3 years of Revelio).
- MSA FE are redundant given Tract $\times$ Year FE — drop from eq. (1) or footnote it.
- Bank-level clustering is right, but if a few large banks drive identification the effective cluster count is small $\Rightarrow$ wild-cluster bootstrap, leave-one-out.
- The IV sample differs: 12.4M (Table 5) vs. 13.9M (Table 2). Which banks drop, and why?
- Mechanism tests (Tables 6, 7) use uninstrumented $Bank\ AI$ while the headline leans on IV for causality. Instrument them too, or soften the language.
- Table 6 nulls: report SEs/CIs and minimum detectable effects. Where does % minority loan officers come from (NMLS + name-based race, a la Frame et al.)?
- Table 6, col. (2): you read one insignificant coefficient for its sign, then treat the next as a zero. Pick one standard.
- Table 7 shows composition *does* shift (significant interaction on CLTV and LTI). That argues against “minorities got riskier,” not against “composition changed.” Minority applicants looking observably *safer* yet denied more is either the finding or selection on unobservables — adding FICO decides which.

Institutions, framing, literature

- **Ginnie Mae is not a GSE.** It guarantees FHA/VA/USDA loans underwritten under those agencies' rules — the DU/LPA standardization story applies to Fannie/Freddie only.
- Split refis into **cash-out vs. rate-and-term**; cash-out underwriting looks much more like purchase.
- The **FinTech list** (Buchak et al. 2018) predates the sample — misclassified FinTech-style nonbanks land in the “bank” sample. Also, that sample includes non-FinTech *nonbank* mortgage companies; consider depositories vs. nonbanks separately.
- **Discount points** include seller/third-party-paid amounts, unlike the other two fee measures. Use the borrower-paid portion.
- **McDash:** ~20% match rate — compare matched vs. unmatched and reweight. Interest rate is both a merge key and the outcome; rerun matching without it.

Notes for the author — Free disposal! (4): framing and literature

- **Disaggregate Minority** into Black and Hispanic (different histories, geographies, FHA shares). Report missing-race rates by *Bank AI*.
- **Magnitudes:** implied extra minority denials per year; dollars of extra interest over loan life at 12 bp for p95 banks.
- **Engage directly:** Bhutta, Hizmo & Ringo (JF 2025); Bhutta & Hizmo (RFS 2021); Bartlett et al. (2022) and Howell et al. (2024) — algorithms/automation *reduce* disparities in their settings. Reconciling your sign is a contribution, not a threat.
- **Regulatory context:** ECOA/Reg B adverse-action rules for complex models (CFPB Circular 2022-03) and SR 11-7 model risk management shape how much ML can sit inside bank mortgage decisions in 2022–24.

Notes for the author — Free disposal! (5): small stuff

- Page 1 (title / authors / abstract) was missing from the PDF I received.
- “AI Job Share” vs. “Bank AI”: pick one name. Same for “AI Strong” vs. “Historical AI hiring” (Table 4 vs. Table 5) — and it matters, because the placebo only validates whichever variable it actually tested.
- “Interest Spread” (APR – Treasury, Table 1/A1) vs. “Rate Spread” (APR – APOR, Table 12). Table 1’s row is labeled *Rate Spread*. Align them.
- §3.2: “negative coefficient estimates on *Loan to Income*. . . By contrast, higher loan-to-income is associated with worse outcomes” reads as self-contradictory.
- §3.4: the IV rate estimate is $\sim 1.8 \times$ OLS ($3.552/2.019$), not “more than double.”
- “Median income $\approx$ \$99,000” is $\exp(p50)$; the mean log (4.627) implies $\approx$ \$102k. Just say “median.”
- “our bank AI adoption measure is available for 2001–2023” looks like a typo (elsewhere: 2021–2023).
- Typos: “three three strands”; “Section 4 rule out”; “disadvantageing”; “e also consistent.”
- “INSERT TABLE X HERE” placeholders still in the draft.
- **Nice-to-have:** β_1 rises from 0.265 (col. 1) to 0.375 (cols. 2–3) as FE get tighter. Worth a sentence — the confounder is not in the direction those FE absorb.